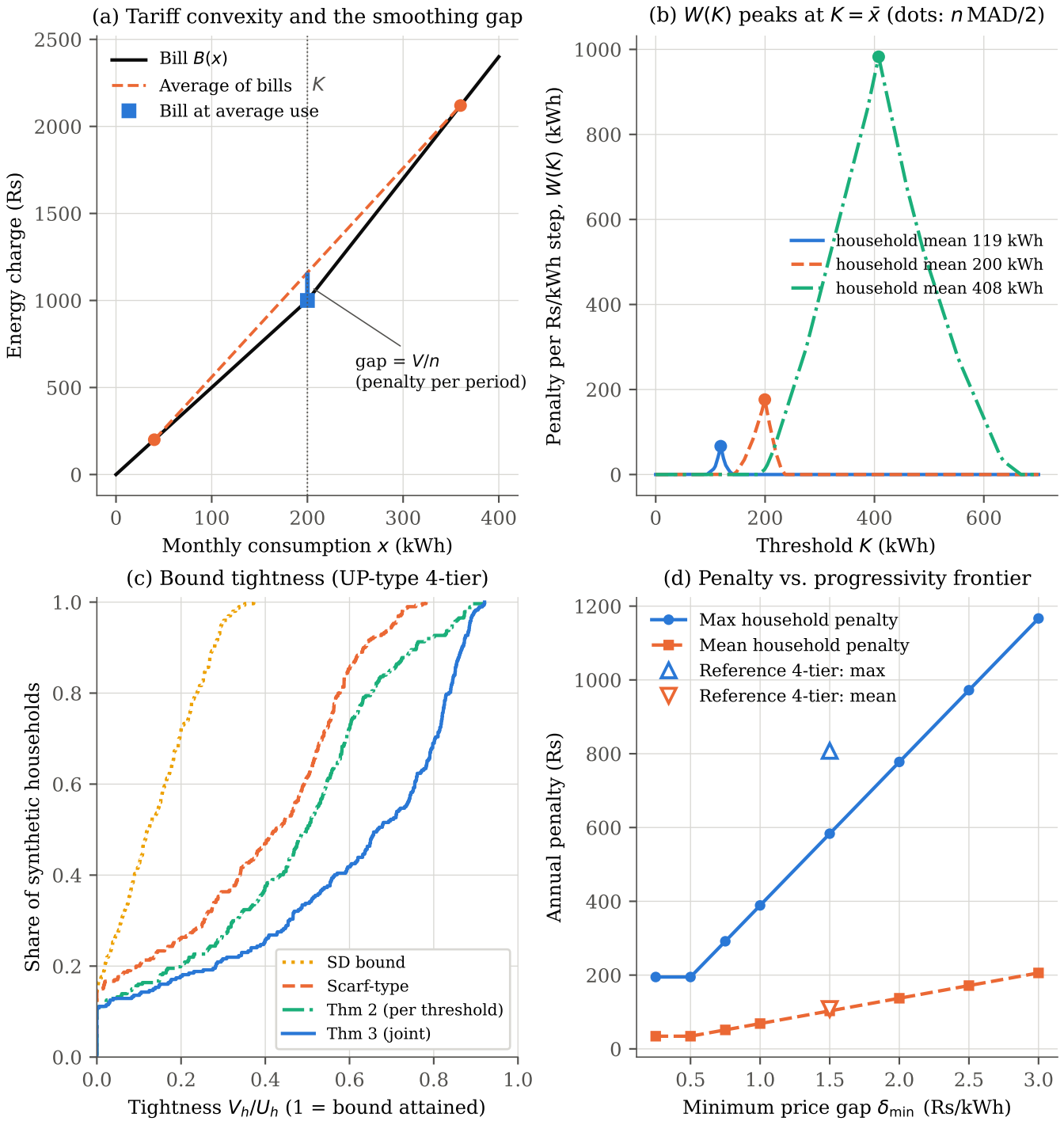
(a) Tariff convexity and the smoothing gap
Bill $B(x)$
Average of bills
Bill at average use
K
gap = V/n
(penalty per period)
Energy charge (Rs)
Monthly consumption x (kWh)
(b) $W(K)$ peaks at $K = \bar{x}$ (dots: n MAD/2)
household mean 119 kWh
household mean 200 kWh
household mean 408 kWh
Penalty per Rs/kWh step, $W(K)$ (kWh)
Threshold K (kWh)
(c) Bound tightness (UP-type 4-tier)
SD bound
Scarf-type
Thm 2 (per threshold)
Thm 3 (joint)
Share of synthetic households
Tightness V_h/U_h (1 = bound attained)
(d) Penalty vs. progressivity frontier
Max household penalty
Mean household penalty
Reference 4-tier: max
Reference 4-tier: mean
Annual penalty (Rs)
Minimum price gap $\delta_{\min}$ (Rs/kWh)